Test utility Mean ± sample SD

Method	Qwen2.5-7B		Llama-3.1-8B		Qwen3.8-27B	
	AppWorld	PlanBench-XL	AppWorld	PlanBench-XL	AppWorld	PlanBench-XL
Frozen Base	0.2583 ±0.0109	0.2633 ±0.0304	0.2720 ±0.0121	0.2682 ±0.0338	0.3341 ±0.0101	0.3290 ±0.0429
AREW / T3	0.2771 ±0.0127	0.2809 ±0.0218	0.2826 ±0.0087	0.3068 ±0.0433	0.3665 ±0.0085	0.3744 ±0.0377
GiGPO	0.3094 ±0.0299	0.3163 ±0.0395	0.3295 ±0.0268	0.3419 ±0.0558	0.3952 ±0.0071	0.4070 ±0.0395
InfoPO	0.3634 ±0.0289	0.3975 ±0.0447	0.3955 ±0.0035	0.4163 ±0.0449	0.5268 ±0.0676	0.5279 ±0.0689
WorldEvolver	0.3636 ±0.0188	0.4441 ±0.0253	0.3827 ±0.0052	0.4541 ±0.0347	0.4487 ±0.0081	0.5227 ±0.0228
PoS / EnvProbe	0.5112 ±0.0021	0.4921 ±0.0234	0.5583 ±0.0059	0.5627 ±0.0224	0.6649 ±0.0104	0.6630 ±0.0081
Step-DPO	0.5583 ±0.0713	0.5356 ±0.0730	**0.5871** ±0.0636	0.5607 ±0.0658	**0.6770** ±0.0398	**0.6643** ±0.0358
Outcome-PG	0.2999 ±0.0141	0.3186 ±0.0356	0.3138 ±0.0147	0.3520 ±0.0704	0.4078 ±0.0332	0.4099 ±0.0353
Outcome-Value	0.3084 ±0.0262	0.3316 ±0.0470	0.3310 ±0.0219	0.3693 ±0.0601	0.4104 ±0.0302	0.4099 ±0.0353
Outcome-RLOO	0.3056 ±0.0224	0.3212 ±0.0311	0.3188 ±0.0266	0.3546 ±0.0660	0.4038 ±0.0384	0.4099 ±0.0353
Info-RL Neighbor	0.3350 ±0.0320	0.3693 ±0.0610	0.3633 ±0.0197	0.3900 ±0.0511	0.4635 ±0.0390	0.4233 ±0.0412
Unstratified	0.3444 ±0.0226	0.3926 ±0.0398	0.3808 ±0.0100	0.3998 ±0.0399	0.4712 ±0.0137	0.4743 ±0.0397
Unweighted	**0.5842** ±0.1161	**0.5932** ±0.0605	0.5279 ±0.1105	**0.5939** ±0.0828	0.6484 ±0.0432	0.6386 ±0.0444
SPAR	**0.6130** ±0.1138	**0.6243** ±0.0470	**0.6279** ±0.1102	**0.6677** ±0.0197	**0.6948** ±0.0091	**0.6847** ±0.0130